Force-free time-harmonic plasmoids: harmonics and size

Nachamkin's ball-lightning plasmoid studied as written, over the reported range of size, energy and lifetime

kronspice simulation report · 2026-11-08 · about 5 seconds (plus 25 s for the pictures); deterministic

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Summary

J. Nachamkin's 1992 Phillips Laboratory report (PL-TR-92-3044) models ball lightning as a sphere of plasma holding a standing electromagnetic wave whose electric and magnetic fields are parallel everywhere (a force-free field), with a vortical flow balancing its pressure and a "resonant size" at which it should not radiate. This example takes the paper as written and asks three questions at ball-lightning scale: which frequencies (harmonics) a plasmoid of each size may have, what the paper's own recipe gives over the reported range of size and energy, and how long such a plasmoid lasts. kronspice analyses its spectrum, its ring-down and its charge condition as circuits.

- **The field is sound.** Eq. 8 is force-free to 1.8×10^{-10}; its energy integral is $35.868/\alpha^3$, as the paper's B3 says.
- **Appendix B has an arithmetic slip.** Its field $E_0 = 3.566\times10^{9}$ V/m follows from $35.87/\alpha^2$, not from its own B3 (α^3): the field should be $\sqrt{\alpha} = 6.7$ times larger. Every other printed number is reproduced.
- **Each size allows a comb of harmonics** $f_n = x_n \cdot c/2\pi R$ (x_n the zeros of j_1); at the mean ball-lightning radius, 12 cm, they start at 3.07 GHz.
- **The paper's recipe drives its electrons faster than light**, $v/c = \xi \cdot kR = 1.4$ to 12 c, over the whole reported range; keeping them below 0.1 c moves the plasmoid into the terahertz.
- **Free, it lasts microseconds to milliseconds, not seconds.** Lasting the 1–10 s that witnesses report needs, at a 12 cm radius, 7–74 free electrons per sea-level air molecule; fully stripped air has 14.5.
- **kronspice** returns the plasmoid's eight lowest resonances to 10^{-15} in frequency and 2×10^{-6} in Q, closes the ring-down ledger to 5×10^{-6}, and shows the paper's charge condition (eq. 49) is the resonance of a plasma-loaded line.

What is being tested

The claim (hypothesis H26)

Long-lived plasmoids (ball lightning) are force-free, time-harmonic electromagnetic fields in a plasma, stabilized by vortical flow and non-radiating at a resonant size: $j_1(\alpha R) = 0$ and $j_1(kR) = 0$ with $k > \alpha$ (eq. 35). The free-electron density is fixed by eq. 49, $\rho_e = (m/e)\varepsilon_0 c^2(k^2 - \alpha^2)$.

The standard prediction beside it

Eq. 49 is the cold-plasma dispersion relation $\omega^2 = \omega_p{}^2 + \alpha^2c^2$. A plasma sphere fed by nothing outside has only leaky (radiating) modes; its TE and TM halves are separate modes needing different sizes, so the force-free mix of the two is not one free mode. Ball lightning as reported: diameter 10–50 cm (mean 24 cm), lifetime peaking at 2–5 s (mean about 10 s), energy 10 kJ to over 1 MJ (eyewitness statistics).

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The plasmoid

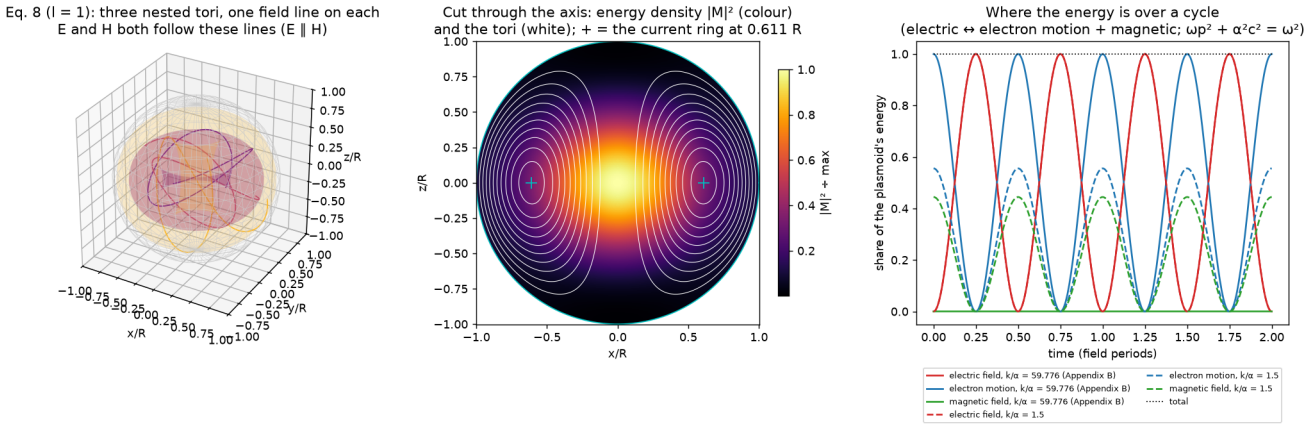


Figure 1. Left: the l = 1 field of eq. 8 lies on nested tori (three shown, one field line on each); E and H follow the same lines. Middle: a cut through the axis, energy density |M|² and the tori, nesting around the current ring at 0.611 R. Right: over a cycle the energy moves between the electric field and the electrons' motion (plus a magnetic share (α/k)², negligible at Appendix B's k/α = 59.8); the total stays constant because $\omega_p^{\ 2} + \alpha^2c^2 = \omega^2$.

Eq. 8 for l = 1: $M = 2\cos\theta \cdot j_1(\alpha r)/(\alpha r)\, e_r - [j_0(\alpha r) - j_1(\alpha r)/(\alpha r)] \sin\theta\, e_\theta \pm j_1(\alpha r) \sin\theta\, e_\varphi$ (unit vectors e), with $E = E_0 M \sin(\omega t + \varphi)$ and H proportional to $M \cos(\omega t + \varphi)$.

Results

1. Appendix B recomputed

Case	E_0 (V/m)	kR used	f (GHz)	ρ_e (C/m³)	Electrons	v/c
As printed, his k = 2686 /m	3.566×10^{9}	268.60	128.16	−32.63	8.532×10^{17}	2.69
As printed, nearest zero	3.566×10^{9}	265.46	126.66	−31.87	8.333×10^{17}	2.65
B3 as printed (α³)	2.39×10^{10}	683.30	326.0	−211.2	5.52×10^{18}	6.83

- Every printed value of the first row is reproduced; eq. 49 equals the cold-plasma dispersion to 4×10^{-17}.
- His step from k = 2642 to 2686 /m skips the nearest zero of j_1 (265.46) for the next one.
- Scaled to 50 μm (his eq. 86, energy proportional to R): 500 J and λ = 1.17 μm as printed, but 4.27×10^{14} electrons against the printed 4.3×10^{13}, and 30 rather than "about forty" times sea-level density.

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2. Harmonics and size

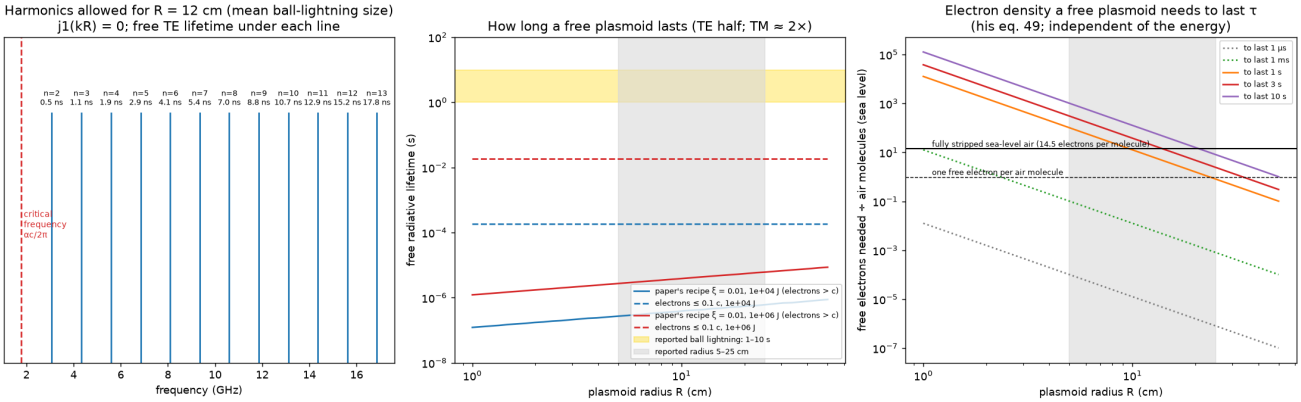


Figure 2. Left: the harmonics eq. 35 allows a 12 cm plasmoid, above the critical frequency αc/2π, with the free TE lifetime of each. Middle: free radiative lifetime against radius, by the paper's recipe (ξ = 0.01; electrons above c) and with the electrons kept at or below 0.1 c, for 10 kJ and 1 MJ, against the reported 1–10 s (gold). Right: the free-electron density a free plasmoid needs to last τ, against fully stripped sea-level air.

Radius	Lifetime wanted	Frequency	Free electrons ÷ sea-level air molecules
5 cm	1 – 10 s	470 – 1485 THz	102 – 1020
12 cm (mean)	1 – 10 s	126 – 399 THz	7.4 – 74
25 cm	1 – 10 s	42 – 133 THz	0.8 – 8.1
50 cm	1 – 10 s	15 – 47 THz	0.10 – 1.0

For kR much larger than 1 the free energy lifetimes are $\tau_{TE} = R(kR)^2/(2c\cdot x_1^2)$ and $\tau_{TM} = R(kR)^2|g'(x_t)|/(2c\cdot x_t^3)$, with $g = 1 + x\cdot j_1'/j_1$, $x_1 = 4.4934$, $x_t = 2.7437$; both agree with the complex root finder to four figures at kR = 268.6, 1000 and 3000. So lasting τ fixes kR, and eq. 49 then fixes the electron density, whatever the energy.

3. kronspice analysis

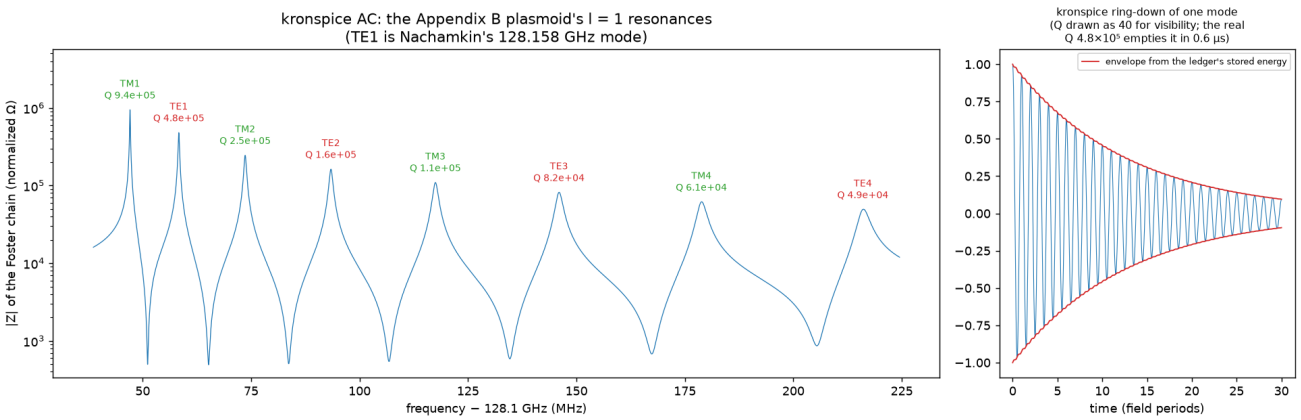


Figure 3. Left: the Appendix B plasmoid's four TE and four TM l = 1 resonances as a Foster chain of parallel RLC tanks in a kronspice AC sweep (TE1 is Nachamkin's 128.158 GHz mode). Right: a kronspice ring-down of one tank, its envelope drawn from the ledger's stored energy.

- Spectrum: each peak's frequency matches the mode finder to ≤ 2×10^{-15} and its half-power Q to 2×10^{-6}.
- Ring-down of TE1 over 400 periods: Q = 4.7992×10^5 against 4.7985×10^5; heat = energy lost to 5×10^{-6} of the stored energy; energy lifetime 0.60 µs.
- Eq. 49 as a circuit: a one-dimensional plasma line (series $\mu_0\Delta$, shunt $\varepsilon_0\Delta$, electrons' kinetic inductance $m/(ne^2\Delta)$ at every node), shorted half a wavelength π/α apart, resonates at $\omega^2 = \omega_p^2 + \alpha^2c^2$: kronspice equals the exact lattice to 7×10^{-10}, and the lattice approaches eq. 49 as $1/N^2$ (7.3×10^{-5} at 50 cells, 4.6×10^{-6} at 200).

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The reported seconds

Witnesses report ball lightning lasting seconds, and this example does not dispute the reports; it measures the paper's model against them. As written, a free plasmoid of the paper's kind radiates its field away in microseconds (the paper's recipe) to tens of milliseconds (electrons kept non-relativistic), at every reported size and energy. Lasting seconds would need more free electrons than the air contains at the mean size. So either the plasmoid is not free, or its energy is not mainly in this field:

- **Held from outside.** The paper itself (§IX) says the boundary must be matched by an external standing field or by surface currents; a plasmoid fed or held that way can last as long as its supply. The microwave-sustained cavity plasmoids the paper mentions are the laboratory case. The test: a measured external field or surface current at the resonant size.
- **Energy stored elsewhere.** Standard models that reach seconds store the energy chemically, not in the field: luminous balls from electrical discharges into silicon last several seconds in the laboratory (Abrahamson & Dinniss, *Nature* 403, 519, 2000; Paiva et al., *Phys. Rev. Lett.* 98, 048501, 2007). These are context, not part of the paper's model.

Ledger and validation

- kronspice ring-down: heat equals the energy lost to 5×10^{-6} of the stored energy.
- Foster-chain AC: frequency to 10^{-15}, Q to 2×10^{-6} for all eight modes.
- Large-kR lifetimes equal the complex root finder (TE to 10^{-3}, TM to 5×10^{-3} in the tests).
- *main_test.go*: field and B3, Appendix B, harmonics, lifetimes, kronspice spectrum, ring-down and plasma line.

Limits of the model

- Lifetimes are for a uniform cold-plasma sphere radiating into free space; the vortical flow, collisions, recombination and any external field are not modelled.
- The energy follows the paper's split (half field, half fluid) and its time-averaged field energy (B2).
- Only l = 1 is analysed in detail, as in the paper; the minimum radii of l = 2–4 are listed.
- Ball-lightning statistics are eyewitness-based and broad.

Status

Register entry H26 stays *untested* as a ball-lightning model. Its field mathematics and charge condition are sound; its worked example has a √α slip and drives electrons faster than light; and a free plasmoid of its kind cannot last the seconds reported at any reported size without more free electrons than air has. What would move it: a measured external field or surface current holding a plasmoid at its resonant size.

Reproducing the results

```
go run ./examples/Force-Free_Harmonic_Plasmoids -viz examples/Force-Free_Harmonic_Plasmoids/viz
python examples/Force-Free_Harmonic_Plasmoids/viz/render.py      # pictures and plasmoid_cycle.gif
python examples/Force-Free_Harmonic_Plasmoids/viz/make_report.py # this report
go test ./examples/Force-Free_Harmonic_Plasmoids
```

Sources: J. Nachamkin, *Force-Free Time-Harmonic Plasmoids*, PL-TR-92-3044 (1992), DTIC AD-A257 765; S. Chandrasekhar, P. C. Kendall, *Astrophys. J.* 126, 457 (1957); J. A. Stratton, *Electromagnetic Theory* (1941), §9.22; F. F. Chen, *Introduction to Plasma Physics*; ball-lightning statistics: arXiv:physics/0312122, arXiv:1102.2829; J. Abrahamson, J. Dinniss, *Nature* 403, 519 (2000); G. S. Paiva et al., *Phys. Rev. Lett.* 98, 048501 (2007).

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